

EMBEDDINGS & FIRST RETRIEVAL

WEEK 3, DAY 1

Turning text into numbers, finding what is relevant,
and learning exactly where that goes wrong.





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PART ONE: MEANING AS NUMBERS

What an embedding is, and why similarity is just geometry

TEXT BECOMES COORDINATES

AN EMBEDDING IS A LIST OF NUMBERS

ONE DOCUMENT, ONE VECTOR

"Okonkwo Bell is a boutique in energy..."

↓

[0.031, -0.118, 0.902, ...] 1024 numbers

SIMILAR MEANING

Vectors point in a similar direction, even with no words in common.

UNRELATED MEANING

Vectors point elsewhere, even if they share plenty of words.

SIMILARITY IS JUST AN ANGLE

COSINE SIMILARITY, IN THREE LINES OF PYTHON

Score	What it means
1.0	Same direction. As close to identical meaning as the model can tell.
0.6 – 0.9	Related. This is where useful retrieval actually lives.
0.0	Unrelated. The vectors are at right angles to each other.
Below your floor	Retrieved, but not relevant. You must decide what to do about this.

Search always returns something. Something is not the same as relevant.



PART TWO: WHERE IT BREAKS

Negation, exact identifiers, rare terms - and knowing when to refuse

THREE THINGS IT CANNOT DO

AND ALL THREE LOOK LIKE SUCCESS

- | | | |
|---|--------------------------|---|
| 1 | Negation | "does not operate in the US" looks almost identical to "operates in the US" |
| 2 | Exact identifiers | Ask for matter LP-2291 and you get documents that feel similar instead |
| 3 | Rare terms | Jargon the model barely saw in training lands nowhere useful |

Every one of these returns a confident, well-ranked, wrong answer.

REFUSING IS A FEATURE

THE THRESHOLD IS YOUR DECISION, NOT THE MODEL'S

WITHOUT A FLOOR

Top three documents go to the model regardless. It writes something plausible from irrelevant context.

WITH A FLOOR

Nothing clears the bar, so you never call the model. You return a refusal and spend nothing.

"I do not have a document that answers that" is a correct answer.

A LIST, THEN A REAL STORE

THE THIRD TIME YOU HAVE DONE THIS

In memory	What it costs you
A Python list	Gone on restart. Re-embed the whole corpus every time you reboot.
Your own cosine loop	Compares against every document. Fine at eight, hopeless at eighty thousand.
Chroma	Persists to disk, indexes properly, same two function names as before.

Build it by hand first. You cannot debug a library you never understood.